

Exercises

Exercise 1

Suppose the electron angular distribution in beta decay is

$$W(\theta) = 1 + a \cos \theta.$$

Show that under spatial inversion,

$$\mathbf{r} \rightarrow -\mathbf{r},$$

the angle transforms as

$$\theta \rightarrow \pi - \theta,$$

and therefore

$$W(\theta) \rightarrow W'(\theta) = 1 - a \cos \theta.$$

Explain why a nonzero coefficient a implies parity violation.